C7 Organic Chemistry – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 1: Crude Oil and Alkanes **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Crude oil				Finite mixture of hydrocarbons from ancient plankton
2	Hydrocarbon				Compound of hydrogen and carbon only
3	General formula of alkanes				C_nH_{2n+2}
4	First four alkanes				Methane, ethane, propane, butane
5	Fractional distillation separates by				Boiling point
6	Fractions from crude oil				LPG, petrol, kerosene, diesel, heavy fuel oil
7	Longer chains: boiling point				Higher
8	Longer chains: viscosity				Higher
9	Longer chains: flammability				Lower
10	Products of complete combustion				Carbon dioxide + water

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C7 Organic Chemistry – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 2: Cracking and Alkenes **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Cracking				Breaking long hydrocarbons into shorter, more useful ones
2	Two methods of cracking				Catalytic cracking and steam cracking
3	Products of cracking				Shorter alkanes and alkenes
4	Why cracking is needed				Short chains are in higher demand
5	General formula of alkenes				C_nH_{2n}
6	Alkene functional group				C=C
7	Test for alkenes				Bromine water
8	Positive result for alkenes				Orange to colourless
9	Uses of alkenes				Making polymers and other chemicals
10	Unsaturated				Contains a C=C double bond

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C7 Organic Chemistry – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 3: Alkenes and Alcohols Paper 2 Triple only

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	First four alkenes				Ethene, propene, butene, pentene
2	Alkene + hydrogen				Alkane (catalyst)
3	Alkene + steam				Alcohol (catalyst)
4	Alkene + bromine				Dibromoalkane
5	Alcohol functional group				–OH
6	First four alcohols				Methanol, ethanol, propanol, butanol
7	Alcohol + sodium				Hydrogen gas given off
8	Product of oxidising an alcohol				Carboxylic acid
9	Conditions for fermentation				Sugar solution, yeast, warm, no oxygen
10	Uses of ethanol				Fuel, solvent, drinks

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C7 Organic Chemistry – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 4: Carboxylic Acids, Esters and Polymers Paper 2 Triple only

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Carboxylic acid functional group				–COOH
2	First four carboxylic acids				Methanoic, ethanoic, propanoic, butanoic
3	Carboxylic acid + carbonate				Salt + water + carbon dioxide
4	Why carboxylic acids are weak acids				Partially ionise
5	Ester is made from				Carboxylic acid + alcohol (acid catalyst)
6	Ester from ethanol and ethanoic acid				Ethyl ethanoate
7	Monomers for addition polymerisation				Alkenes
8	Condensation polymerisation				Monomers with two functional groups join, losing a small molecule
9	Small molecule usually lost				Water
10	Monomers for a polyester				Diol + dicarboxylic acid

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C7 Organic Chemistry – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 5: Natural Polymers **Paper 2** **Triple only**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Functional groups in an amino acid				Amine ($-NH_2$) and carboxylic acid (–COOH)
2	Amino acids polymerise to form				Polypeptides and proteins
3	DNA stands for				Deoxyribonucleic acid
4	Monomers of DNA				Nucleotides
5	Structure of DNA				Double helix
6	Examples of natural polymers				Proteins, starch, cellulose, DNA
7	Monomer of starch and cellulose				Glucose
8	Type of polymerisation in proteins				Condensation
9	Polymer from ethene				Poly(ethene)
10	Functional groups needed per monomer (condensation)				2

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.